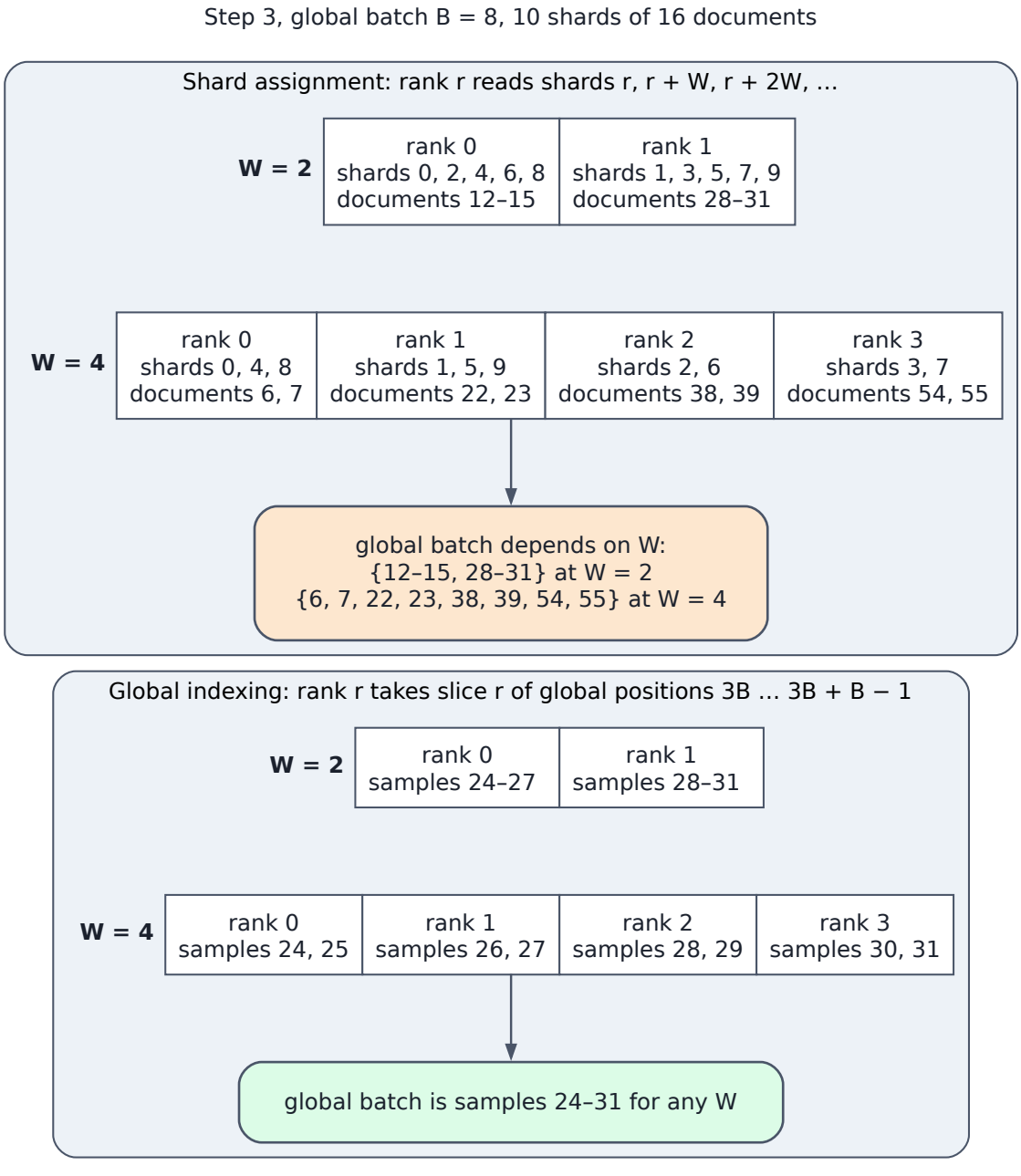
Step 3, global batch B = 8, 10 shards of 16 documents
Shard assignment: rank r reads shards r, r + W, r + 2W, ...
W = 2
rank 0
shards 0, 2, 4, 6, 8
documents 12–15
rank 1
shards 1, 3, 5, 7, 9
documents 28–31
W = 4
rank 0
shards 0, 4, 8
documents 6, 7
rank 1
shards 1, 5, 9
documents 22, 23
rank 2
shards 2, 6
documents 38, 39
rank 3
shards 3, 7
documents 54, 55
global batch depends on W:
{12–15, 28–31} at W = 2
{6, 7, 22, 23, 38, 39, 54, 55} at W = 4
Global indexing: rank r takes slice r of global positions 3B ... 3B + B − 1
W = 2
rank 0
samples 24–27
rank 1
samples 28–31
W = 4
rank 0
samples 24, 25
rank 1
samples 26, 27
rank 2
samples 28, 29
rank 3
samples 30, 31
global batch is samples 24–31 for any W